NATIONAL MATHEMATICS CONTEST
O-LEVEL PAPER 2 SOLUTIONS.
SATURDAY 29TH JUNE, 2024

SECTION A: 5 marks each

1. Let x, y, z and w be integers satisfying $x\log_{10}(2) + y\log_{10}(3) + z\log_{10}(5) + w\log_{10}(7) = 2024$. Use the above information to find the value of $4x + 3y + 2z + w$.

Solution:

Exponentiating the expression $x\log_{10}(2)+y\log_{10}(3)+z\log_{10}(5)+w\log_{10}(7) = 2024$ yields $2^x3^y5^z7^w = 10^{2024} = 2^{2024}3^05^{2024}7^0$. By the fundamental theorem of arithmetic, (compare terms on both sides) $x = z = 2024$ and $y = w = 0$, hence $4x + 3y + 2z + w = 12144$.

2. Find the number of arrangements of $4, 5, 6, 8, 9, 10, 14$ into a single row so that every pair of the adjacent numbers in the arrangement has a common factor greater than 1.

Solution:

Let $X = \{4, 5, 6, 8, 9, 10, 14\}$. The arrangement must either start with $5, 10$ and end with $6, 9$ or start with $9, 6$ and end with $10, 5$ since $gcd(5, x) = 1$ for every $x \in X$ except when $x = 10$, and $gcd(9, x) = 1$ for every $x \in X$ except when $x = 6$.
Since the greatest common factor of $\{4, 6, 8, 10, 14\}$ is 2, then every permutation of $4, 8, 14$ between the 6 and 10 gives a possible arrangement, thus there are $2 \times 3! = 12$ ways and these are;

$5, 10, 14, 8, 4, 6, 9$	$5, 10, 14, 4, 8, 6, 9$
$5, 10, 8, 14, 4, 6, 9$	$5, 10, 8, 4, 14, 6, 9$
$5, 10, 4, 8, 14, 6, 9$	$5, 10, 4, 14, 8, 6, 9$
$9, 6, 14, 8, 4, 10, 5$	$9, 6, 14, 4, 8, 10, 5$
$9, 6, 8, 14, 4, 10, 5$	$9, 6, 8, 4, 14, 10, 5$
$9, 6, 4, 8, 14, 10, 5$	$9, 6, 4, 14, 8, 10, 5$

3. Let $f : \mathbb{R}_{\geqslant 0} \to \mathbb{R}_{\geqslant 0}$, $x \mapsto f(x)$ be a function such that $f(x+1) = \dfrac{3}{1 + \frac{1}{f(x)}}$. Determine $f(1)$ if $f(7) = \dfrac{4374}{2191}$.

Solution:

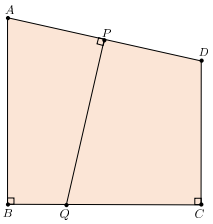
From $f(x+1) = \dfrac{3}{1 + \frac{1}{f(x)}}$, we see that $f(x) \neq 0$ and $f(x) = \dfrac{f(x+1)}{3 - f(x+1)}$. In particular, $f(7) = \dfrac{4374}{2191}$ implies that $f(6) = \dfrac{1458}{733}$, $f(5) = \dfrac{486}{247}$, $f(4) = \dfrac{162}{85}$, $f(3) = \dfrac{54}{31}$, $f(2) = \dfrac{18}{13}$, hence $f(1) = \dfrac{6}{7}$. Algorithmically, $f(6) = \dfrac{\frac{4374}{2191}}{3 - \frac{4374}{2191}} = \dfrac{\frac{4374}{3}}{2191 - \frac{4374}{3}} = \dfrac{1458}{733}$.

4. Two friends, Anzoa and Origa play a game in which Anzoa rolls two ordinary dice while Origa rolls one. What is the probability that Origa's die roll exceeds the total of Anzoa's?

Solution:

The set of Origa's possible outcomes is $\{x \in \mathbb{Z} : 1 \leqslant x \leqslant 6\}$ while that of Anzoa is $\{(x, y) \in \mathbb{Z}^2 : 1 \leqslant x, y \leqslant 6\}$. Furthermore, the sum of Anzoa's output belongs to the set $\{z \in \mathbb{Z} : 2 \leqslant z \leqslant 12\}$. Let X be the random variable for Origa's output and Z be the random variable for the sum of Anzoa's out put. Depending on whether Origa rolls 1, 2, 3, 4, 5, or 6, the probability that Anzoa's total is lower is respectively $0, 0, \frac{1}{36}, \frac{3}{36}, \frac{6}{36}, \frac{10}{36}$. Since these outputs of Anzoa's and Origa's rolls are independent, it follows from the multiplicative rule that, the overall probability that Origa's die roll exceeds the total of Anzoa's is $\sum_{i=1}^{6} \Pr(\{X = i\} \cap \{Z \geqslant i\}) = \sum_{i=1}^{6} \Pr(\{X = i\}) \cdot \Pr(\{Z < i\}) = \frac{1}{6}\left(\frac{1}{36} + \frac{3}{36} + \frac{6}{36} + \frac{10}{36}\right) = \frac{5}{54}$.

5. In the Figure on the right, $\overrightarrow{\texttt{AB}}$ is perpendicular to $\overrightarrow{\texttt{BC}}$, $\overrightarrow{\texttt{DC}}$ is perpendicular to $\overrightarrow{\texttt{BC}}$, and $\overrightarrow{\texttt{PQ}}$ is the perpendicular bisector of $\overrightarrow{\texttt{AD}}$. If $\overrightarrow{\texttt{AB}} = 9$, $\overrightarrow{\texttt{BC}} = 8$, and $\overrightarrow{\texttt{DC}} = 7$, find the area of the quadrilateral APQB.



Solution:

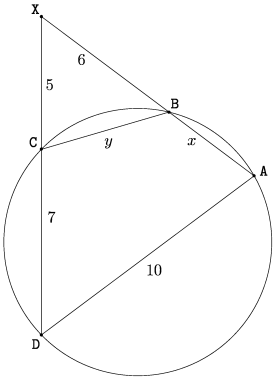
Shift the origin of the xy-plane to be at B. Since P is the midpoint of $\overrightarrow{\texttt{AD}}$, its coordinates are $(4, 8)$. Let (a, b) be the coordinates of Q, the intersection of the perpendicular bisector of $\overrightarrow{\texttt{AD}}$ and the line segment $\overrightarrow{\texttt{BC}}$. Since along $\overrightarrow{\texttt{AD}}$, is along the x-axis, we have $b = 0$. By gradient considerations, we have $\frac{0-8}{a-4} \cdot \frac{7-9}{8-0} = -1$, hence $a = 2$. Therefore, the area enclosed by the quadrilateral APQB is $\frac{1}{2} \cdot 8(2 + 4) + \frac{1}{2} \cdot 1 \cdot 4 = 26$ square units.

6. There are 2024 antelopes and a certain number of lions living across nine game parks in Uganda. In each park, the number of antelopes is eleven times the total number of lions in all the other parks. Find the number of lions that live in the nine game parks of Uganda.

Solution:

Let $X_1, \ldots, X_9$ be the number of antelopes in each of the nine parks and let $x_1, \ldots, x_9$ be the number of lions in each of the nine parks so that a park contains X antelopes and x lions and so on. Therefore, $X_1 = 11(x_2 + x_3 + \cdots + x_9)$ and $X_2 = 11(x_1 + x_3 + \cdots + x_9)$ and so on. Therefore, $X_1 + \cdots + X_9 = 8 \cdot 11(x_1 + \cdots + x_9)$ and hence, since total number of antelopes is 2024, we have $2024 = 88(x_1 + \cdots + x_9)$. Therefore $(x_1 + \cdots + x_9) = \frac{2024}{88} = 23$ and hence the total number of lions in the nine parks is 23.

7. A cyclic quadrilateral has all four vertices on the circumference of a circle. Find the area enclosed by the cyclic quadrilateral ABCD with sides $x, y, 7, 10$ shown in the figure on the right. (In 598 - 670 AD, Brahmagupta showed the area A, enclosed by a cyclic quadrilateral whose edges have lengths a, b, c, d to be



$$A := \sqrt{(s-a)(s-b)(s-c)(s-d)}$$

where $s = \dfrac{a+b+c+d}{2}$).

Solution:

From the Figure, triangles XDA and XBC are similar, hence, $\dfrac{7+5}{6} = \dfrac{10}{y} = \dfrac{6+x}{5}$, thence $x = 4$ and $y = 5$. Let $a = 4$, $b = 5$, $c = 7$ and $d = 10$ and A be the plane area enclosed by a cyclic quadrilateral with edge lengths a, b, c, d. From this, we obtain $s = \dfrac{a+b+c+d}{2} = 13$ hence $A = \sqrt{(s-a)(s-b)(s-c)(s-d)} = 36$ square units by Brahmagupta's formula.

8. Suppose a, b, and c are positive integers with $a < b < c$ such that $\dfrac{1}{a} + \dfrac{1}{b} + \dfrac{1}{c} = 1$. What is $a+b+c$?

Solution:

First note that we must have $\dfrac{1}{a} < 1$, so $a > 1$. Since $\dfrac{1}{a} > \dfrac{1}{b} > \dfrac{1}{c}$, we must also have $\dfrac{1}{a} > \dfrac{1}{3}$; so $a < 3$. Thus, $a = 2$. Now $\dfrac{1}{b} + \dfrac{1}{c} = \dfrac{1}{2}$ where $2 < b < c$. Similar to before, $\dfrac{1}{b} > \dfrac{1}{4}$, so $b < 4$. Thus, $b = 3$. With $a = 2$ and $b = 3$ we have $\dfrac{1}{2} + \dfrac{1}{3} + \dfrac{1}{c} = 1$, which is satisfied when $c = 6$. To conclude,

$$a + b + c = 2 + 3 + 6 = 11.$$

9. A house is in the shape of a triangle, perimeter P metres and area A square metres. The garden consists of all the land within 5 metres of the house. How much land the garden and the house together occupy?

Solution:

The garden consists of 3 rectangular pieces and three sectors of a circle. The rectangular pieces all have width 5 metres and the total length is P metres. Their combined area is therefore $5P$ square metres.

At a corner, with interior angle x, the angle within the sector is $360° - 180° - x = 180° - x$. The sum of the angles in all three sectors is $3(180°) -$ (sum of interior angles) $= 360°$.

Therefore, the sectors fit together to form a circle of radius 5, and the combined area is 25π. The total area of house and garden is $A + 25\pi + 5p$ square metres.

10. Find the sum of all the even integers from 546 to 854 inclusive.

Solution:

$$(546 + 854) + (548 + 852) + (550 + 850) + \cdots + (696 + 704) + (698 + 702) + 700$$
$$1400 + 1400 + 1400 + \cdots + 1400 + 1400 + 700$$

Since $698 - 546 = 152$ and we have only even numbers we have $76 + 1 = 77$ times 1400 are added together.

$$(77 \times 1400) + 700 = 108500$$

SECTION B: 10 marks each

11. A Mersenne prime is any prime number of the form $2^\ell - 1$, where $\ell \in \mathbb{Z}_{\geqslant 2}$, for example, 3 and 7. Show that, if an integer n is the sum of two different powers of 2 and also the sum of two different Mersenne primes, then n is the sum of two different square numbers. For example, $10 = 2^3 + 2 = (2^3 - 1) + (2^2 - 1) = 7 + 3 = 3^2 + 1^2$. **Note:** $\mathbb{Z}_{\geqslant 2}$ is a set of positive integers greater or equal to 2.

Solution:

Suppose n is the sum of two different powers of 2 and also the sum of two different Mersenne primes, i.e., $n = 2^x + 2^y$ for some $x, y \in \mathbb{Z}_{\geqslant 0}$ with $x \neq y$ and $n = (2^w - 1) + (2^z - 1)$ where $(2^w - 1)$ and $(2^z - 1)$ are primes with $w \neq z$. Without loss of generality, suppose $x > y$ and $w > z$. Since all Mersenne primes are odd, it follows that n is even hence $y \geqslant 1$. Furthermore, $2^x + 2^y = n = 2^w + 2^z - 2$ implies that $2^{x-1} + 2^{y-1} = 2^{w-1} + 2^{z-1} - 1$. Since $z \geqslant 2$, it follows that $2^{w-1} + 2^{z-1} - 1$ is odd hence $y = 1$ in which case we find that $2^{x-2} = 2^{w-2} + 2^{z-2} - 1$. Now, if $z > 2$, then there are no positive integer solutions. However, if $z = 2$, then $x = w$ (which must be an odd prime number since $w > z = 2$) and so $n = 2^x + 2$. In this case, we have $x = 2r + 1$ for some $r \in \mathbb{Z}_{\geqslant 1}$ and $n = 2^{2r+1} + 2 = (2^r + 1)^2 + (2^r - 1)^2$.

12. Let ABCD be a cyclic quadrilateral such that the line segments $\overrightarrow{CD}$ and $\overrightarrow{BA}$ are extended to meet at E. Furthermore, suppose that the (straight) line through D which is tangent to the circle ADE meets the (straight) line $\overrightarrow{CB}$ at F. Prove that the triangle CDF is isosceles.

Solution:

There are two possible diagrams based on the description in the question, though one is given in Figure 1. In the cyclic quadrilateral ABCD, we have, $\angle DCB + \angle DAB = \pi$ (opposite interior angles of a cyclic quadrilateral are supplementary) and $\angle EAD + \angle DAB = \pi$, (angles on straightline are supplementary). Therefore, $\angle EAD = \angle DCF$, (or by the alternate segment theorem). Using the alternate segment theorem with circle ADE and tangent at D, we deduce that $\angle EAD = \angle FDC$ (vertically opposite angles), hence $\angle FDC = \angle FCD$. Using the properties of isosceles triangles, we can conclude that triangle CDF is isosceles.

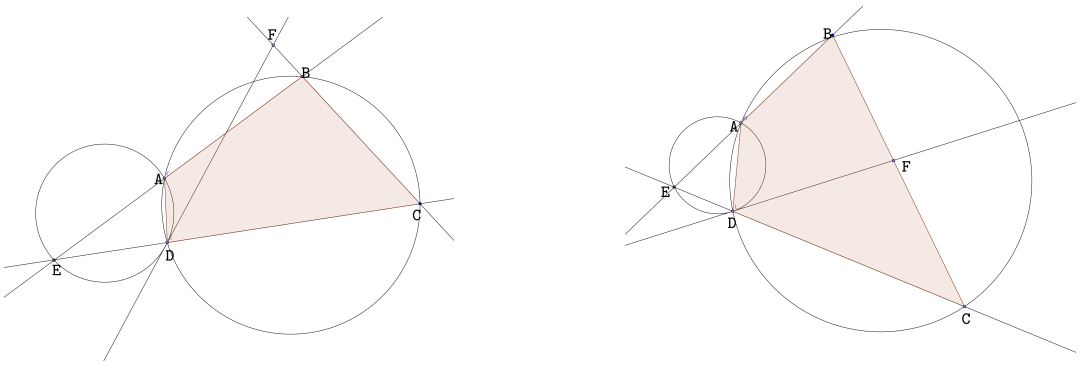
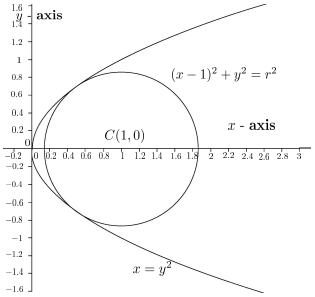


Figure 1: Sketch diagram of the first configuration.

13. A parabola $x = y^2$ intersects (in at least one point) the circle with center $\mathtt{C}(1,0)$ and radius $r > 0$ (has equation $(x-1)^2+y^2 = r^2$). Find the least value of r for which this is true.



Solution:

Let (α, β) be the point of intersection of the parabola $x = y^2$ and the circle centered at $(1,0)$ of radius r. It is clear that, $\alpha = \beta^2$ and $r = \sqrt{(\alpha-1)^2+(\beta-0)^2} = \sqrt{(\beta^2-1)^2+\beta^2}$. By completion of squares, $r = \sqrt{(\beta^2-1)^2+\beta^2} = \sqrt{\beta^4-\beta^2+1} = \sqrt{\left(\beta^2-\frac{1}{2}\right)^2+\frac{3}{4}} \geqslant \frac{\sqrt{3}}{2}$.

14. If x, y, and z are positive real numbers satisfying $x^2+y^2 = 9$, $y^2+yz+z^2 = 16$ and $x^2+\sqrt{3}xz+z^2 = 25$, what is the value of $2xy+xz+\sqrt{3}yz$?

Solution:

We need to find the value of $2xy+xz+\sqrt{3}yz$

$$x^2+y^2 = 9 \quad (1.1)$$
$$y^2+yz+z^2 = 16 \quad (1.2)$$
$$x^2+\sqrt{3}xz+z^2 = 25 \quad (1.3)$$

from $(x+y)^2 = x^2+2xy+y^2$, we have $(x+y)^2-2yx = x^2+y^2$ substitute into Eqn (1.1) to have $(x+y)^2-2xy = 9$ that imply

$$2xy = (x+y)^2-9 \quad (1.4)$$

from $(y+z)^2 = y^2+2yz+z^2$, we have $(y+z)^2-yz = y^2+zy+z^2$, substitute into Eqn (1.2) to have $(y+z)^2-yz = 16 \Rightarrow yz = (y+z)^2-16$

$$\sqrt{3}yz = \sqrt{3}(y+z)^2-16\sqrt{3} \quad (1.5)$$

from $(x+z)^2 = x^2 + 2x + z^2$, we have

$$(x+z)^2 - 2xz = x^2 + z^2 \tag{1.6}$$

adding equations (1.1) to (1.2) we get,

$$x^2 + 2y^2 + yz + z^2 = 25 \tag{1.7}$$

Equations (1.4) = (1.3) so that $x^2 + \sqrt{3}xz + z^2 = x^2 + 2y^2 + yz + z^2$ to have

$$\sqrt{3}xz = 2y^2 + yz \tag{1.8}$$

Add equations (1.6) to (1.8) to get,
$x^2+\sqrt{3}xz+z^2 = (x+z)^2-2xz+2y^2+yz$ (substitute into (1.3)) to have $25 = (x+z)^2-2xz+2y^2+yz$. Therefore,

$$xz = \frac{1}{2}\left((x+z)^2 + 2y^2 + yz - 25\right) \tag{1.9}$$

Adding (1.4), (1.5) and (1.9), we get

$$2xy + xz + \sqrt{3}yz = (x+y)^2 - 9 + \sqrt{3}(y+z)^2 - 16\sqrt{3} + \frac{1}{2}((x+z)^2 + 2y^2 + yz - 25)$$

15. Peter and John played a game. Peter wrote on a blackboard all integers from 1 to 18 and offered John to choose 8 different integers from this list. To win the game John had to choose 8 integers such that among them the difference between any two is either less than 7 or greater than 11. Can John win the game? Justify your answer.

Solution:

Assume that such integers $\{x_1, x_2, x_3, x_4, x_5, x_6, x_7, x_8\}$ are chosen and listed in the increasing order. Since all requirements are about differences between integers, we can shift them and assume that $x_1 = 1$. Since the difference between any two integers is either less than 7 or greater than 11, the integers $8, 9, 10, 11, 12$ cannot be in the list. Consider the following pairs of the remaining integers: $(2, 13)$, $(3, 14)$, $(4, 15)$, $(5, 16)$, $(6, 17)$, $(7, 18)$. The difference between two integers, forming the same pair, is 11. Thus, only one integer from each pair can be represented in the list. So these 6 pairs can produce only six integers in the list. With $x_1 = 1$ it makes 7. Therefore, 8 integers cannot be chosen.

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